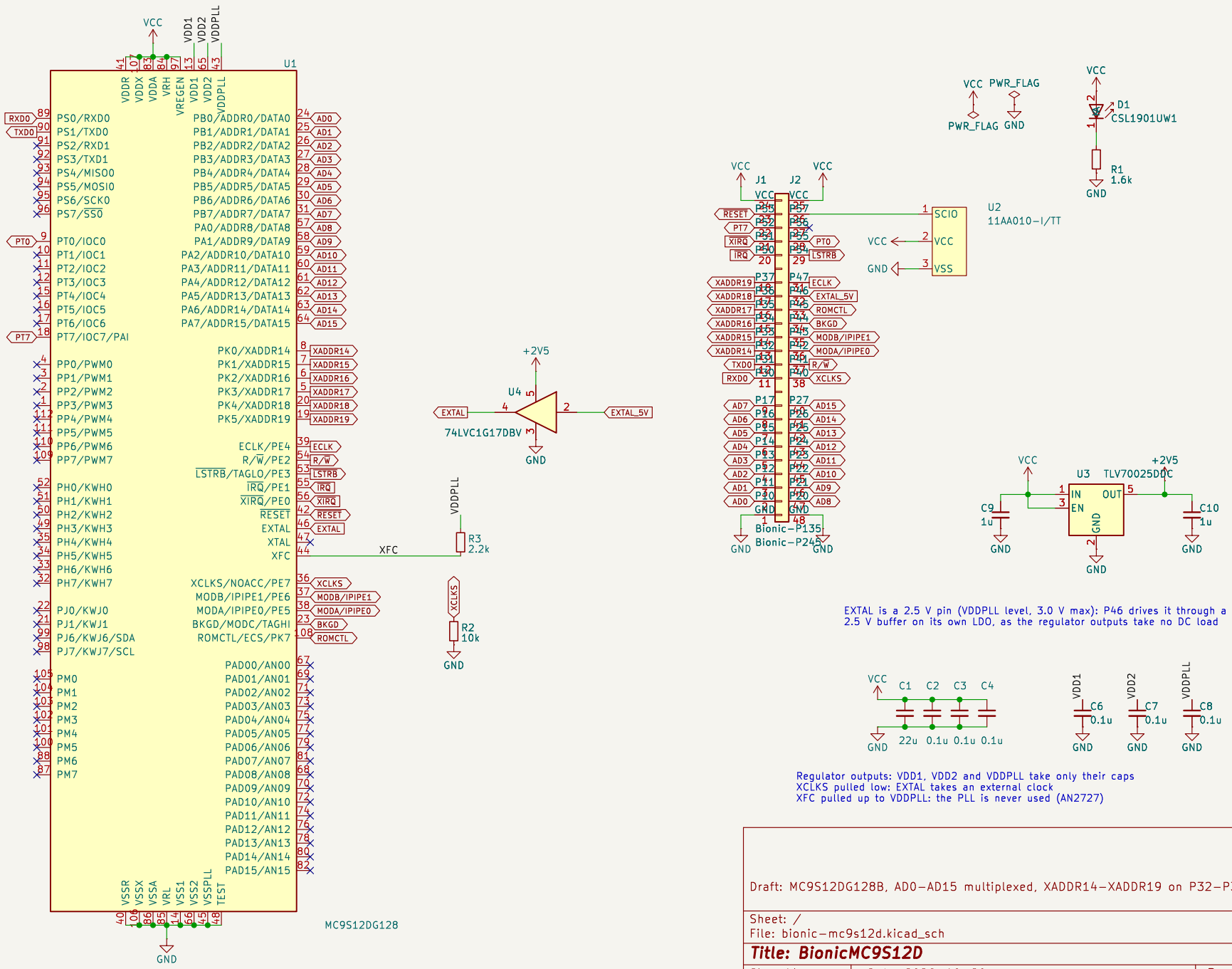

U1
MC9S12DG128
PS0/RXD0
PS1/TXD0
PS2/RXD1
PS3/TXD1
PS4/MISO0
PS5/MOSI0
PS6/SCK0
PS7/SS0
PT0/IOC0
PT1/IOC1
PT2/IOC2
PT3/IOC3
PT4/IOC4
PT5/IOC5
PT6/IOC6
PT7/IOC7/PAI
PP0/PWM0
PP1/PWM1
PP2/PWM2
PP3/PWM3
PP4/PWM4
PP5/PWM5
PP6/PWM6
PP7/PWM7
PH0/KWH0
PH1/KWH1
PH2/KWH2
PH3/KWH3
PH4/KWH4
PH5/KWH5
PH6/KWH6
PH7/KWH7
PJ0/KWJ0
PJ1/KWJ1
PJ6/KWJ6/SDA
PJ7/KWJ7/SCL
PM0
PM1
PM2
PM3
PM4
PM5
PM6
PM7
PB0/ADDR0/DATA0
PB1/ADDR1/DATA1
PB2/ADDR2/DATA2
PB3/ADDR3/DATA3
PB4/ADDR4/DATA4
PB5/ADDR5/DATA5
PB6/ADDR6/DATA6
PB7/ADDR7/DATA7
PA0/ADDR8/DATA8
PA1/ADDR9/DATA9
PA2/ADDR10/DATA10
PA3/ADDR11/DATA11
PA4/ADDR12/DATA12
PA5/ADDR13/DATA13
PA6/ADDR14/DATA14
PA7/ADDR15/DATA15
PK0/XADDR14
PK1/XADDR15
PK2/XADDR16
PK3/XADDR17
PK4/XADDR18
PK5/XADDR19
ECLK/PE4
R/W/PE2
LSTRB/TAGLO/PE3
IRQ/PE1
XIRQ/PE0
RESET
EXTAL
XTAL
XFC
XCLKS/NOACC/PE7
MODB/IPIPE1/PE6
MODA/IPIPE0/PE5
BKGD/MODC/TAGHI
ROMCTL/ECS/PK7
PAD00/AN00
PAD01/AN01
PAD02/AN02
PAD03/AN03
PAD04/AN04
PAD05/AN05
PAD06/AN06
PAD07/AN07
PAD08/AN08
PAD09/AN09
PAD10/AN10
PAD11/AN11
PAD12/AN12
PAD13/AN13
PAD14/AN14
PAD15/AN15
VCC
VDD1
VDD2
VDDPLL
VDDR
VDDX
VDDA
VRH
VREGEN
VSSR
VSSX
VSSA
VRL
VSS1
VSS2
VSSPLL
TEST
GND
U4
74LVC1G17DBV
+2V5
EXTAL_5V
R3
2.2k
R2
10k
J1
J2
Bionic−P135
Bionic−P245
U2
11AA010−I/TT
SCIO
U3
TLV70025DDC
IN
OUT
EN
C9
1u
C10
1u
D1
CSL1901UW1
R1
1.6k
PWR_FLAG
C1
C2
C3
C4
22u
0.1u
C6
C7
C8
EXTAL is a 2.5 V pin (VDDPLL level, 3.0 V max): P46 drives it through a 2.5 V buffer on its own LDO, as the regulator outputs take no DC load
Regulator outputs: VDD1, VDD2 and VDDPLL take only their caps
XCLKS pulled low: EXTAL takes an external clock
XFC pulled up to VDDPLL: the PLL is never used (AN2727)
Draft: MC9S12DG128B, AD0−AD15 multiplexed, XADDR14−XADDR19 on P32−P37
Sheet: /
File: bionic−mc9s12d.kicad_sch
Title: BionicMC9S12D
Size: A4
Date: 2026−10−20
Rev: 2
KiCad E.D.A. 9.0.8
Id: 1/1